

# ATOMIC STRUCTURE – ATOMS, ELEMENTS & COMPOUNDS

## KEY VOCABULARY

**Element:** a substance made of only one type of atom.

**Compound:** two or more elements chemically bonded together.

**Mixture:** substances together but not chemically bonded.

**Atom:** the smallest particle of an element.

## COMMON MISCONCEPTIONS

~~A compound is just a mixture of elements.~~

✓ A compound is chemically bonded; a mixture is not.

~~Compounds keep the properties of their elements.~~

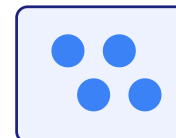
✓ A compound has new, different properties.

~~All substances are made of molecules.~~

✓ Many are made of atoms or ions, not molecules.

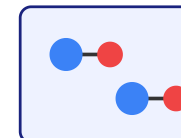
## ELEMENT · COMPOUND · MIXTURE

ELEMENT



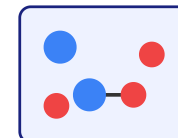
one type of atom

COMPOUND



atoms chemically bonded

MIXTURE



not chemically bonded

A **compound** can only be split by a chemical reaction; a **mixture** can be separated physically.

### 1 DEFINE A COMPOUND

State what is meant by a **compound**. [1 mark]

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### 2 CLASSIFY SUBSTANCES

Classify each as element, compound or mixture: O<sub>2</sub>, H<sub>2</sub>O, air. [3 marks]

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### 3 FORMULAE

State the number of atoms of each element in CaCO<sub>3</sub>. [1 mark]

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### 4 COMPOUND VS MIXTURE

Explain one difference between a compound and a mixture. [2 marks]

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### 5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

An element contains \_\_\_\_ type of atom. The elements in a compound are joined \_\_\_\_.

one

chemically

physically

two

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### 6 FORMING A COMPOUND

Iron and sulfur are heated together. Explain why the product is a compound, not a mixture. [2 marks]

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